Direct Preference Optimization: From Paraphrase Detection to Shakespearean Sonnet Generation

Florencio Paucar

Department of Computer Science, Stanford University

Problem

Aligning language models with human preferences has become a central challenge in NLP. While **Reinforcement Learning from Human Feedback (RLHF)** achieves strong results, its complexity, requiring simultaneous management of multiple models and costly online sampling, motivates simpler alternatives [1].

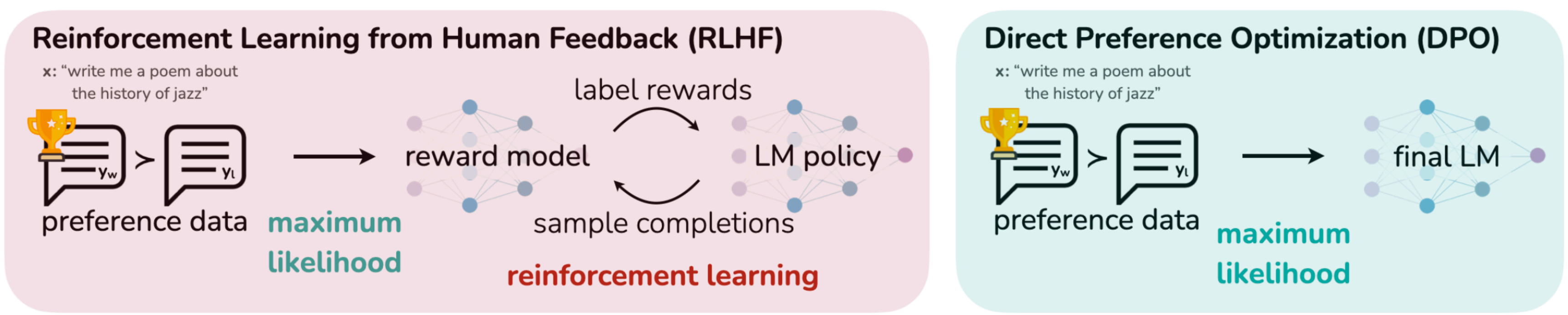


Figure 1. DPO optimizes for human preferences while avoiding reinforcement learning [2].

Direct Preference Optimization (DPO) addresses this by reformulating alignment as a classification problem, eliminating the need for an explicit reward model. However, DPO's effectiveness has been evaluated almost exclusively on open-ended generation tasks.

Goal: This work presents a comprehensive empirical evaluation of DPO applied to two underexplored tasks: paraphrase detection and Shakespearean sonnet generation, both using fine-tuned GPT-2 base models as starting points.

Background

These tasks introduce distinct challenges:

- **Paraphrase detection** offers limited room for improvement given strong supervised baselines.
- **Sonnet generation** demands strict structural and stylistic adherence, which is inherently subjective to evaluate.

DPO Formulation

DPO optimizes the policy directly on preference-labeled datasets using a binary cross-entropy loss:

$$L_{DPO}(\pi_\theta;\pi_{ref}) = -\mathbb{E}_{(x,y_w,y_l)\sim D}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w\mid x)}{\pi_{ref}(y_w\mid x)} - \beta\log\frac{\pi_\theta(y_l\mid x)}{\pi_{ref}(y_l\mid x)}\right)\right]$$

Where:

- σ is the sigmoid function
- π_θ represents the model being optimized
- π_{ref} is a fixed pretrained or fine-tuned base model
- β is a temperature parameter controlling the strength of the preference signal
- (x, y_w, y_l) represents a prompt and its preferred (y_w)/dispreferred (y_l) output pair

Previous work

- Prior sonnet work (Tian Peng et al. 2022; Agnew et al. 2023; Walsh et al. 2024) uses heuristic evaluation or human scoring, no DPO post-training pipeline.
- Prior DPO work on paraphrases (Lübbers et al. 2025; Chen et al. 2026) targets generation, not detection.
- This work delivers one of the first empirical evaluations of DPO on two unexplored tasks: Paraphrase detection and Shakespearean sonnet generation.

Methods

The research methodology is organized into a sequential two phases starting from a GPT-2 base model.

Phase I: Supervised Fine-Tuning (SFT)

1. Full-parameter fine-tuning on the Quora Question Pairs dataset for paraphrase detection.
2. Causal language modeling fine-tuning on a Shakespearean sonnets dataset.

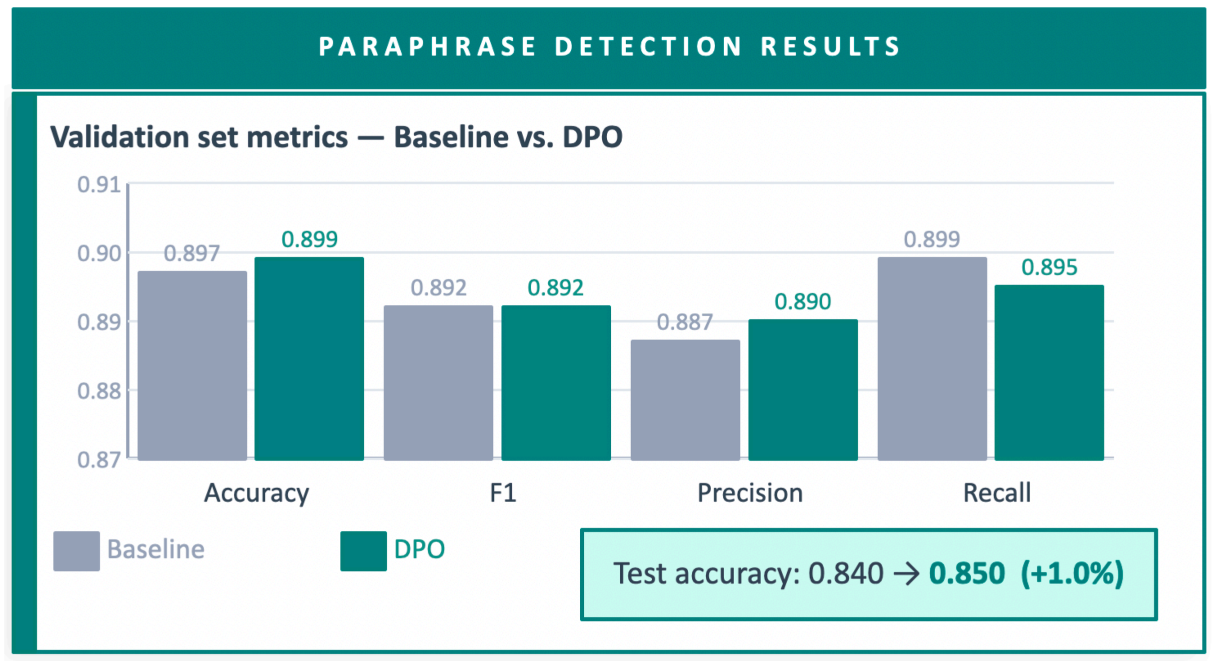
Phase II: Direct Preference Optimization (DPO)

- **Data Preparation**: Gemini 2.5 Flash acted as a high-capacity teacher model to generate preference labels for both tasks.
- **Model Optimization**: Fine-tune the initial SFT policy π_θ by minimizing the L_{DPO} loss function, keeping the starting checkpoint as the reference model π_{ref} to prevent distributional drift.

DPO Experiments

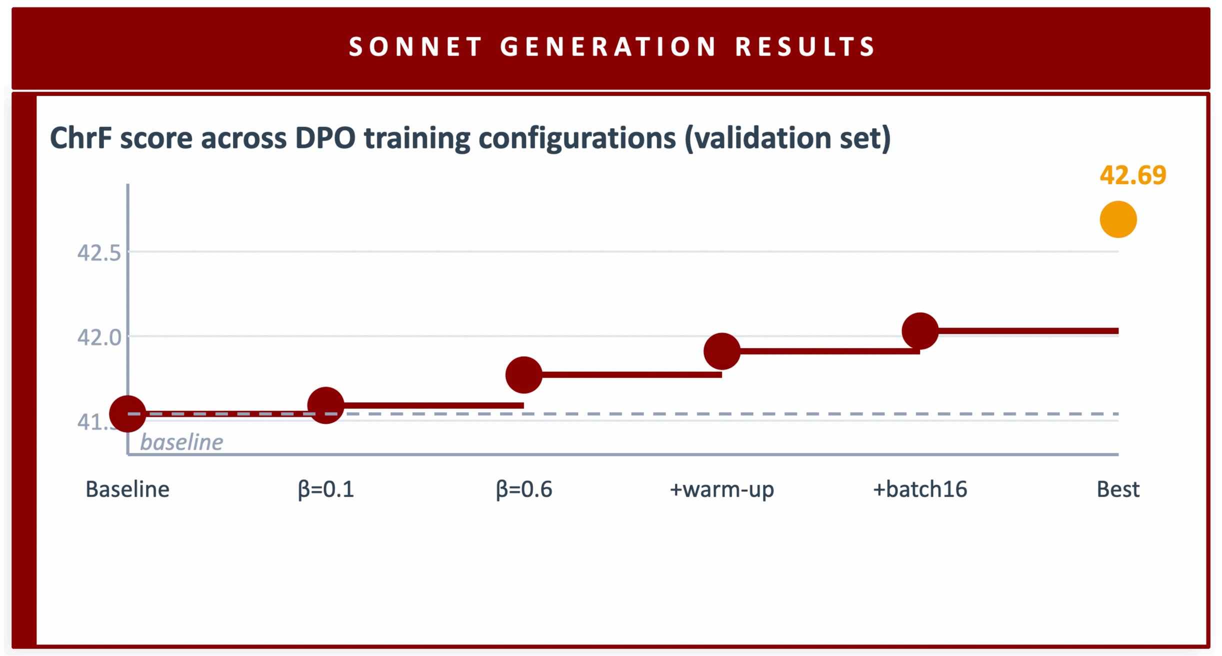
Paraphrase Identification

- **Inputs**: Chosen/rejected paraphrase pairs. **Outputs**: Post-trained Paraphrase classification model.
- **Results**: Consistent, modest improvements.



Sonnet Generation

- **Inputs**: Chosen/rejected sonnets. **Outputs**: Post-trained Sonnet generation model.
- **Results**: **chrF** scores showed mixed results. Slightly decreased from 41.890 to 41.622 on the test set.



Analysis

Sonnets Generation: Text produced after applying DPO is **substantially better** across multiple dimensions invisible to automatic n-gram metrics like chrF.

QUALITATIVE ANALYSIS — SONNET OUTPUT COMPARISON

Dimension	Baseline GPT-2	After DPO ✓
14-line structure	Frequent count errors; occasional catastrophic collapse	Consistently hits 14 lines; structure never collapses
Rhyme scheme	No ABAB CDCD EFEF GG; entirely reliant on Shakespeare's own opening lines	Genuine slant-rhyme attempts in generated continuations
Invented words	19 noise tokens (muzhed, fulshed, murliving, Whunters)	12 invented words; several are plausible archaisms (craftive, meake)
Semantic coherence	Loses thematic thread within 3–4 lines in nearly every sonnet	Sustains original theme for 6–9 lines before drifting

Paraphrase Identification: Our deep dive into edge cases revealed semantic ambiguity on labels. Often human annotators struggle on these boundary cases and relatively strong baseline performance explains the constrained mathematical improvement ceiling post-SFT.

Conclusions

- DPO brings measurable, observable value beyond its originally demonstrated open-ended text domains, proving effective for constrained creative tasks and offering marginal gains in discriminative classification.
- Our sonnet generation findings expose a critical gap between automatic evaluations (e.g. chrF) and genuine stylistic quality in text generation. The true improvements of preference alignment are often structurally profound but formally hard to quantify.
- GPT-2's small capacity caps gains; preference data was generated by a single teacher model, introducing potential bias toward Geminis stylistic preferences.
- Future: Larger models. Iterative DPO sonnet-specific evaluation metrics incorporating rhyme, meter and volta.

References

[1] Paul Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences, 2023.

[2] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. 2024.

fpaucar@stanford.edu